A-A-50156
March 24, 1988

# COMMERCIAL ITEM DESCRIPTION

## HEATERS, HOT WATER BOOSTER, ELECTRIC; SINK, SANITIZING

The General Services Administration has authorized the use of this commercial item description in preference to MIL-H-43895 except for Naval Shipboard use.

## 1. CLASSIFICATION

1.1 Classification. This commercial item description covers hot water booster heaters with the following electrical characteristics (see 5.2):

| Power rating and phase | Voltage |
|---|---|
| 3 Kilowatts, 1 phase | 208, 240, or 480 volts |
| 4.5 Kilowatts, 1 phase | 208, 240, or 480 volts |
| 6 Kilowatts, 3 phase | 208, 240, or 480 volts |
| 9 Kilowatts, 3 phase | 208, 240, or 480 volts |

1.2 Part identification system. The part identification number shall consist of the commercial item description number and the appropriate numbering sequence as follows:

AAXXXXX- X X

- Voltage:
  - A - 208 volt
  - B - 240 volt
  - C - 480 volt
- Power rating and phase:
  - 1 = 3 kilowatts, 1 phase
  - 2 = 4.5 kilowatts, 1 phase
  - 3 = 6 kilowatts, 3 phase
  - 4 = 9 kilowatts, 3 phase
- Commercial item description number

## 2. SALIENT CHARACTERISTICS

2.1 Design and construction. The heaters shall be designed and constructed to be securely mounted to the underside of a sink. The heater shall be complete with all electrical wiring and plumbing. Plumbing connections necessary for attaching the heater to the underside of a sink shall also be provided, and when installation is complete, the overall height of connections shall not exceed 3 inches. When connected to a sink and operated in accordance with the manufacturer's instructions, the heaters shall not show any evidence of leakage. Heater dimensions, exclusive of plumbing fittings, shall not be greater than 18 inches long, 10 inches high, and 8 inches wide.

FSC 7320

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.2 Codes and standards. The heaters shall comply with the applicable requirements of UL 174 or UL 499.

2.3 Heater body and plumbing fittings. The heater body and plumbing fittings shall be fabricated from stainless steel.

2.4 Discharge valve. A quick opening discharge valve shall be provided for draining the water in the sinks. The valve shall be rated for 200°F temperature and shall be readily accessible to the operator when facing the front of the heater.

2.5 ON-OFF switch, temperature indicator, and signal lights. The heater shall have an on-off switch, a temperature indicator to monitor sink temperature, a signal light to indicate when the heater is in operation, and a second signal light to indicate when the high temperature cutout limit has been exceeded. Switch, temperature indicator, and lights shall be appropriately marked.

2.6 Thermostat and high temperature cutout. The heater shall control the sink water temperature between 180°F and 195°F. The heater design shall be such that the thermostat can be set to a temperature within the above limits from the front of the heater. A high temperature cutout shall also be provided to cut power to the heater(s) and prevent heater damage in the event there is no water in the heating reservoir or the regular thermostat fails to operate. The high temperature cutout shall be of the manual reset type.

2.7 Operation. Heaters shall raise the water temperature in sinks with volumes specified below from 140°F to 180°F and maintain a minimum of 180°F for 4 hours:

| Heater rating | Volume of water to be heated |
|---|---|
| 3 kw | Up to 5,000 $in^3$ |
| 4.5 kw | 5,001 to 6,500 $in^3$ |
| 6 kw | 6,501 to 11,500 $in^3$ |
| 9 kw | 11,500 to 18,000 $in^3$ |

2.8 Data name plate (Government unique requirement see 5.4). The heater shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data name plate shall be made of minimum 20 gauge corrosion-resisting metal and attached to the heater by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:



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National stock number
Procurement contract number
Specification data
Manufacturer's name, address and telephone number
Supplier's name, address and telephone number (list only if different from manufacturer)
Manufacturer's model number
Government approved manual number (when specified see 5.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the heater.

2.8.1 Precaution plate. A precaution plate shall be fastened to the heater in a readily visible location with the following information in legible characters:

"WARNING"

"180°F WATER TEMPERATURE"

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify that the product offered meets the salient characteristics and requirements of the description and except for those government unique requirements specified, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. The contractor shall certify the product meets the "Government unique" requirements except when specific quality assurance provisions are specified for the Government unique requirements. In this case, the contractor shall provide data resulting from inspection conducted in accordance with the specified quality assurance. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each item shall be cleaned, preserved, packaged, packed, and unitized in accordance with ASTM D 3951.

4.3 Export packaging. Each heater shall be cleaned, preserved, and cushioned to protect against corrosion, deterioration, and physical damage during shipping and storage.



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4.3.1 Export packing. Each heater shall be packed in a cleated plywood box conforming to overseas type, style optional, type 2 load of PPP-B-601. Each box shall be waterproofed with a sealed case liner conforming to type I, II, or IV, grade optional, of MIL-L-10547. The box shall be closed and strapped in accordance with PPP-B-601. When specified (see 5.2), shipping containers shall be palletized in accordance with MIL-STD-147.

4.4 Marking (commercial and export). Marking of shipping container and unit load shall be in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The heaters are intended for use in kitchen areas for the final rinse sink of a three or four compartment sink washing operation. The following may be used as a guide in selecting a heater with the proper rating for a given size sink:

| Heater (power rating) | Sink size |
|---|---|
| 3 kw | Under 20 gallons |
| 4.5 kw | 20 to 30 gallons |
| 6 kw | 31 to 50 gallons |
| 9 kw | 51 to 80 gallons |

5.2 Ordering data.

a. Title, number, and date of this document.
b. Power rating, phase, and voltage required (see 1.1).
c. Government approved manual number to be included on data name plate when manuals are purchased under the contract (see 2.8).
d. Selection of applicable packaging and packing (see 4.1).
e. When palletization is required (see 4.3.1).

5.3 Sources for nongovernment association documents.

ASTM D 3951 - Standard Practice for Commercial Packaging - Copies are available from: The American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Underwriters Laboratories Standards UL 174 and UL 499 - Copies are available from: The Underwriters Laboratories, Inc., 333 Pfingsten Rd., Northbrook, IL 60062.

5.4 Government unique requirements. Whenever "(Government unique requirements)" is included in the title of a paragraph under "Salient characteristics" it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.



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MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review Activities

DLA - GS
Army - TS, MD
Navy - SA
Air Force - 84

User Activity

Navy - YD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project No. 7320-0829



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